Advanced Machine Learning Algorithms

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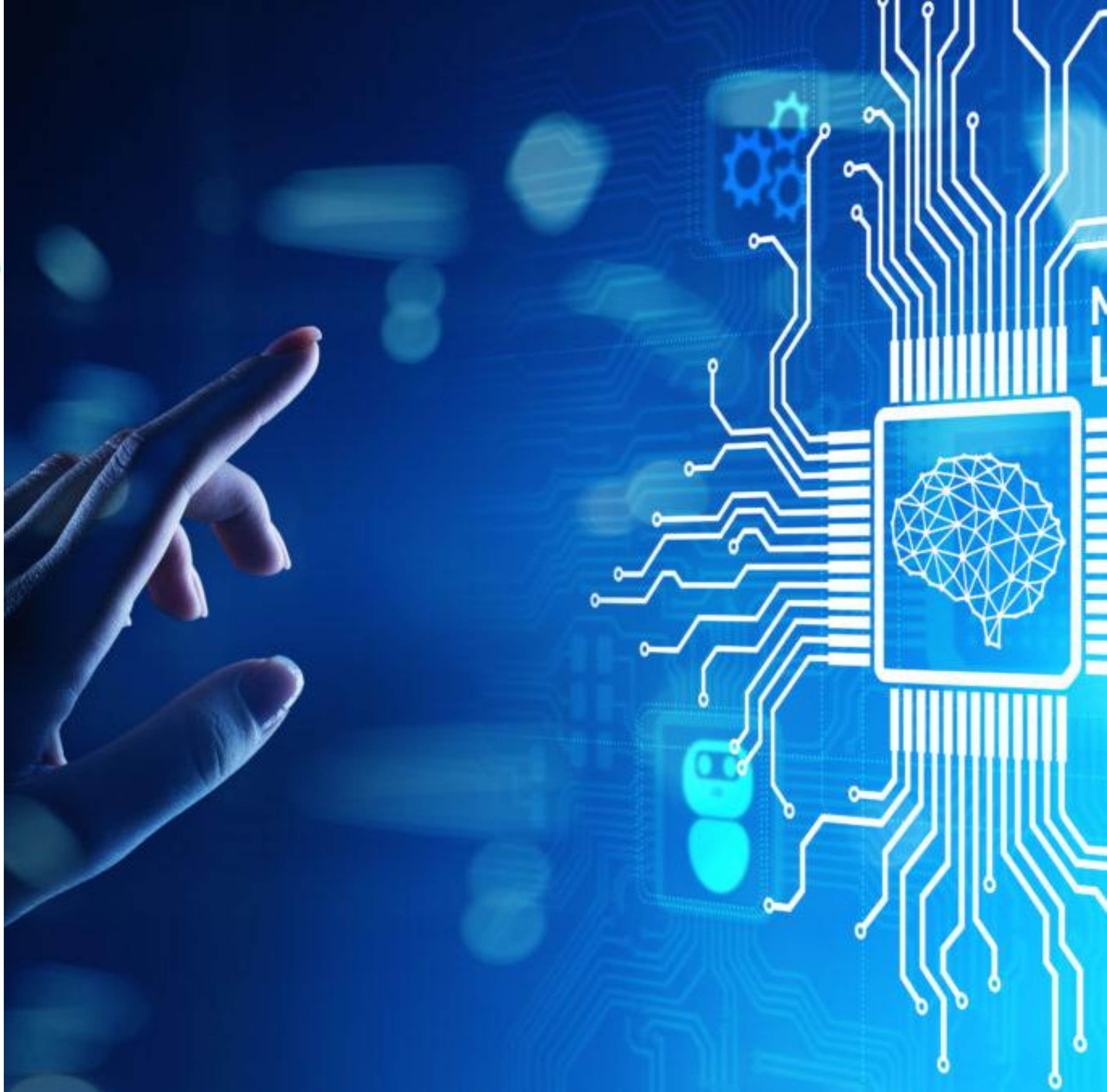

Agenda: Day 1

9 AM – 12 PM:

1. Machine Learning Types
2. Theories of Machine Learning Algorithms
3. Advanced Algorithms : Naïve Bayes

1 PM – 17 PM:

1. Advanced Algorithms : Decision Tree
2. Advanced Algorithms : Random Forest
3. Lab 1: using different Models to solve an advanced Classification Problem.

Agenda: Day 2

9 AM – 12 PM:

1. Neural Networks
2. Feature Engineering
3. Lab 2 : Evaluate and Improve implemented Models with Feature Engineering

1 PM – 17 PM:

1. Evaluation Metrics – Advanced Understanding
2. Hyperparameters Tuning
3. Best Practices to build a performant ML Model
4. Lab 3: Evaluate and Improve implemented Models with Hyperparameter Tuning

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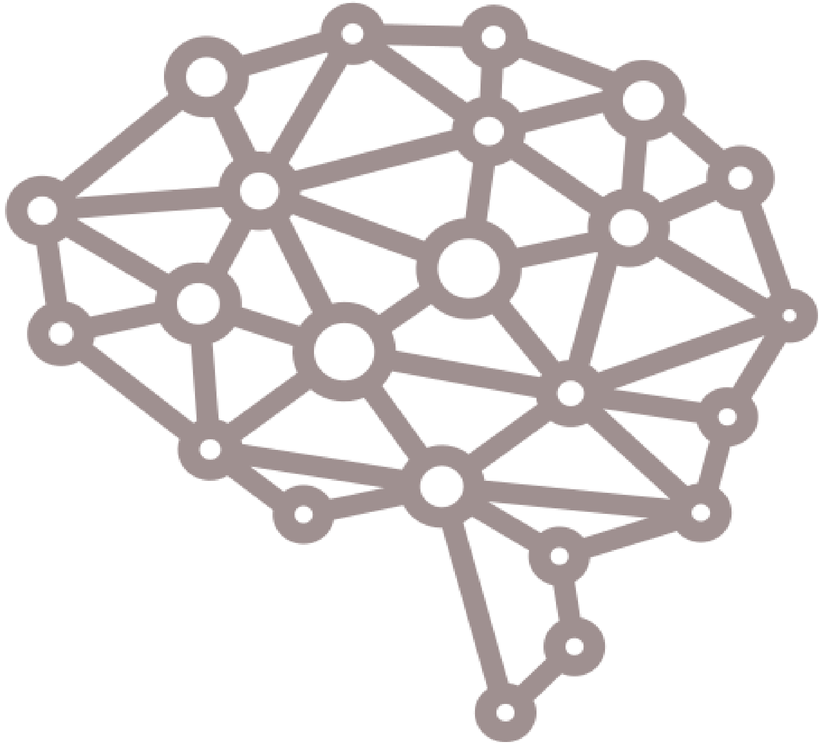
What do you know about neural networks ?



Neural Networks : Popular informations

Introduction to Neural Networks

- Neural Networks started in the 1940s but became really popular in the late 1980s because scientists found better ways to make them and our computers got faster.
- Some models copy how brains work, but not all of them.
- Scientists wanted them to do smart things like brains do. They learn from examples, like kids learning to recognize things, and can figure out stuff even if they haven't seen it before.

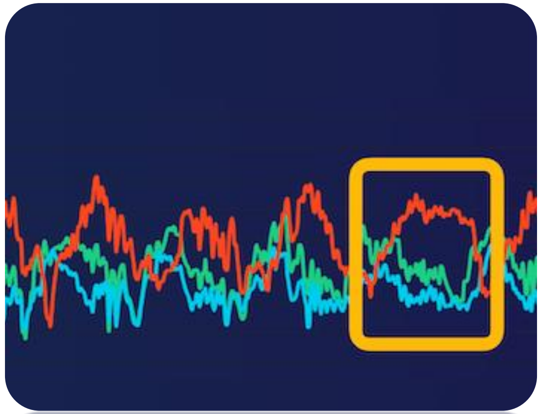
Neural Networks Techniques - learning from examples

- No requirement of an explicit description of the problem.
- No need of a programmer.
- Learns from similar examples during training, even without specific solutions for each problem.
- After sufficient training, it can provide viable solutions for new problems.
- Able to generalize or to handle incomplete data.

Neural Networks : Applications

Pattern recognition

Signal processing



Control systems

Neural Networks vs Computers

Digital computers	Neural Networks
• **Deductive Reasoning.** We apply known rules to input data to produce output. • Computation is centralized, synchronous, and serial. • Memory is packetted, literally stored • Not fault tolerant. One transistor goes and it no longer works. • Exact. • Static connectivity	• **Inductive Reasoning.** Given input and output data (training examples), we construct the rules. • Computation is collective, asynchronous, and parallel. • Memory is distributed, internalized • Fault tolerant, redundancy, and sharing of responsibilities. • Inexact. • Dynamic connectivity.

Applicable if well defined rules with precise input data.

Applicable if rules are unknown or complicated, or if data is noisy or partial.

Neural Networks : Structure of the Brain

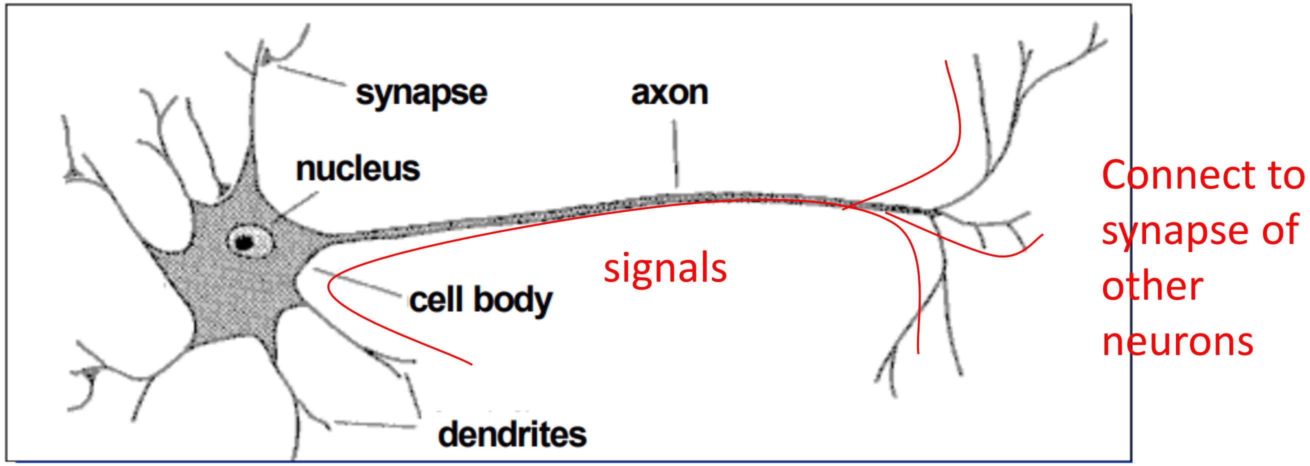
Scientists realized that the brain could solve many problems much easier than even the best computer (image recognition, speech recognition and pattern recognition) so, they studied the brain.

- Each neuron in the brain has a relatively one simple function. But 10 billion of them (60 trillion connections) act together to create an incredible processing unit.
- The brain is trained by its environment and learns by experience
- Even the behaviour of one neuron is extremely complex.
- Engineers modified the neural models to make them more useful less like biology but kept much of the terminology

Structure of Brain Neurons and terminology

A brain neuron has a cell body, an input structure called **dendrite**, and an output structure called **axon**.

Signals are propagated from **dendrites** to the **cell body** and down the **axon** which connect with other neurons through their **synapses**.

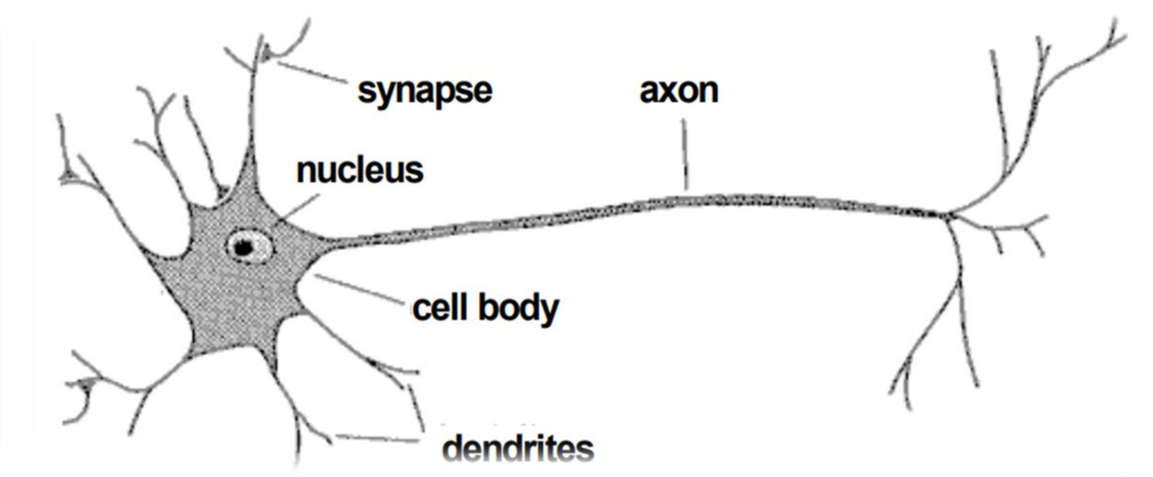


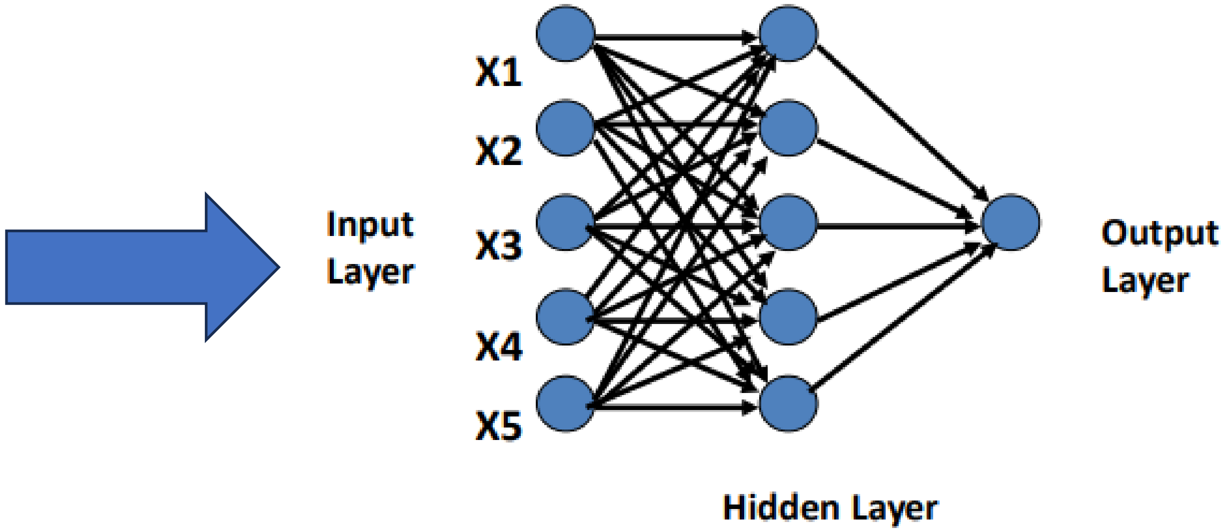
Artificial Neural Networks (ANN) : Key Concepts

- ✓ Scientists has established an equivalent architecture to the brain neural network.

- ✓ This architecture contains :
    - An Input Layer (which replaces the dendrites)
    - A hidden layer where the ANN prepare the responses (which replaces the cell body)
    - An Output layer that replaces the axon.

From Human Brain Neural Network To Artificial Neural Network

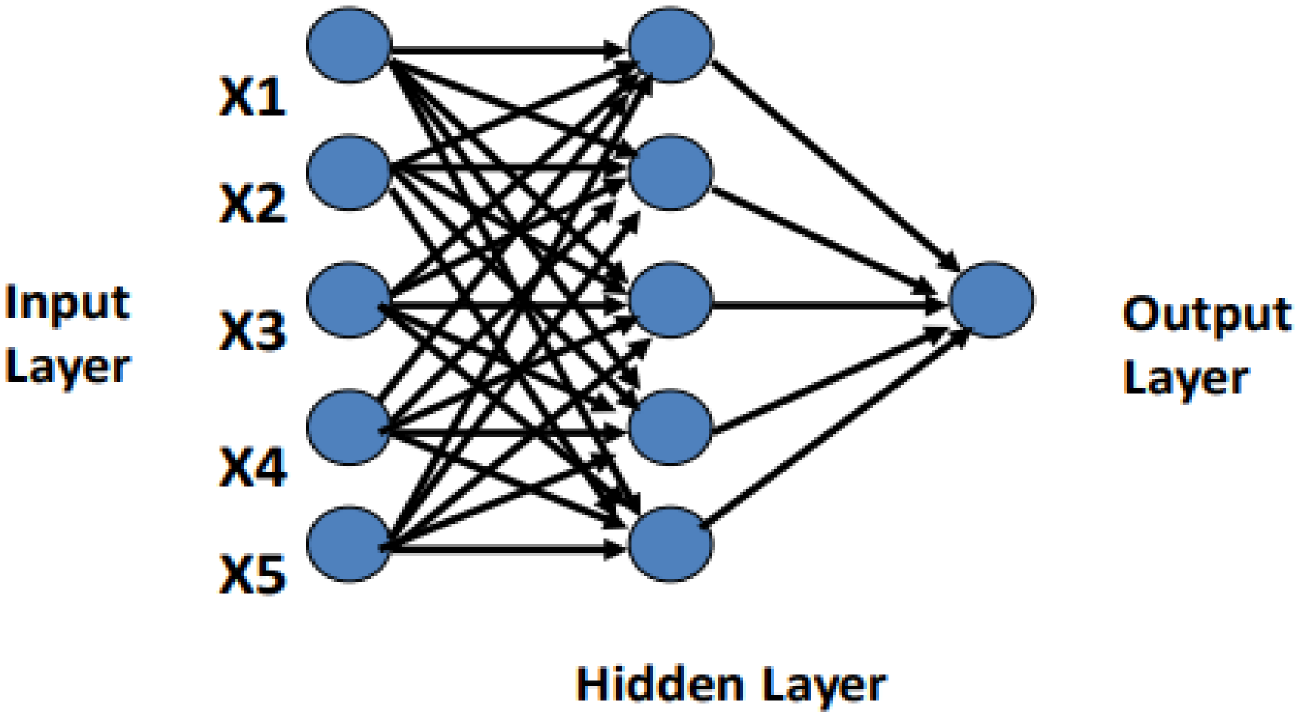




Artificial Neural Networks (ANN) : Architecture

Types of layers

- Input layer
  - Introduces input values into the network
  - No computation or any other processing
- Hidden layer(s)
  - Perform classification of features
  - Two hidden layers are sufficient to solve any problem
- Output layer.
  - Works just like the hidden layers
  - Outputs are passed on to the world outside the neural network



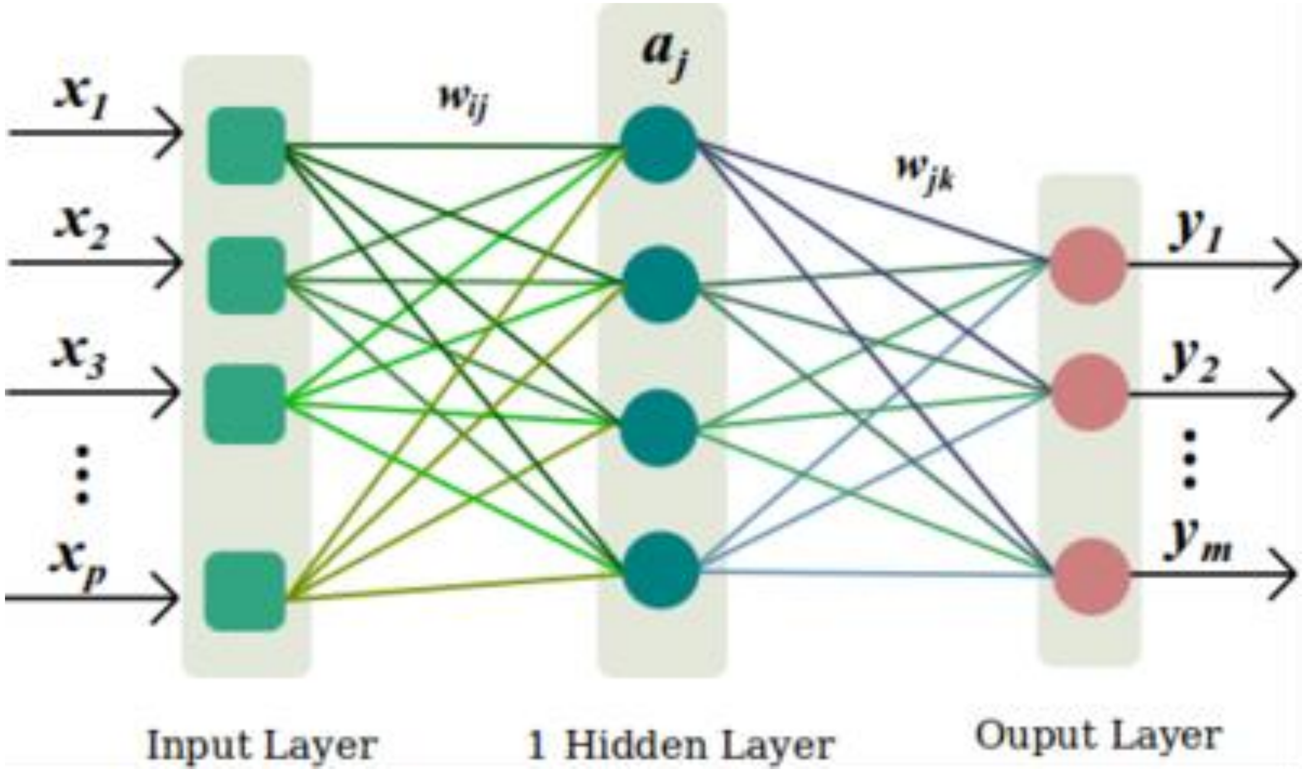
Artificial Neural Networks (ANN) : Architecture

Some key terms to understand the hidden layers :

Weights: Each connection between neurons has a weight, indicating how strong the connection is. If the weight is close to zero, changing the input won't affect the output much. Negative weights mean that increasing the input will decrease the output.

Activation Function: Each neuron in the hidden and output layers applies a function to the sum of its inputs multiplied by their weights. This function is called an activation function. Choosing the right activation function is important and affects how the network learns.

Deep Network: A neural network with two or more hidden layers is called a deep neural network.



Neural Networks : Main Types

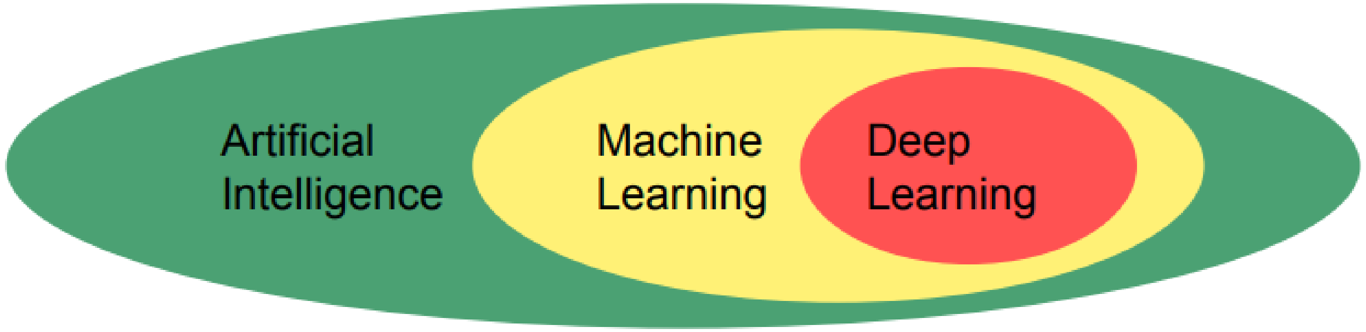
Types of Neural Network	Applications
Feed-Forward Neural Network (FNN)	• Image Recognition • Speech Recognition • Pattern Recognition
Convolutional Neural Network (CNN)	• Image Recognition • Object Detection • Medical Image Analysis
Recurrent Neural Network (RNN)	• Time Series Prediction • Text Processing • Speech Recognition

Deep Learning : Definitions

« Deep learning allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction ».

Deep Learning by Y. LeCun et al. Nature 2015

- ✓ Deep learning is **a branch of machine learning** which is based on artificial neural networks.



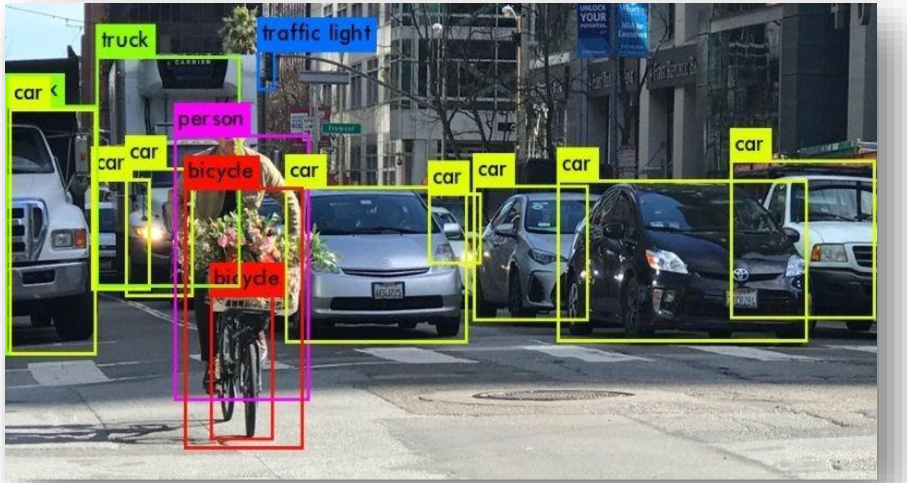
- ✓ The key characteristic of Deep Learning is the use of deep neural networks, which have **multiple layers of interconnected nodes**. Deep Learning algorithms can automatically **learn and improve** from data without the need for manual feature engineering.

Deep Learning vs Machine Learning

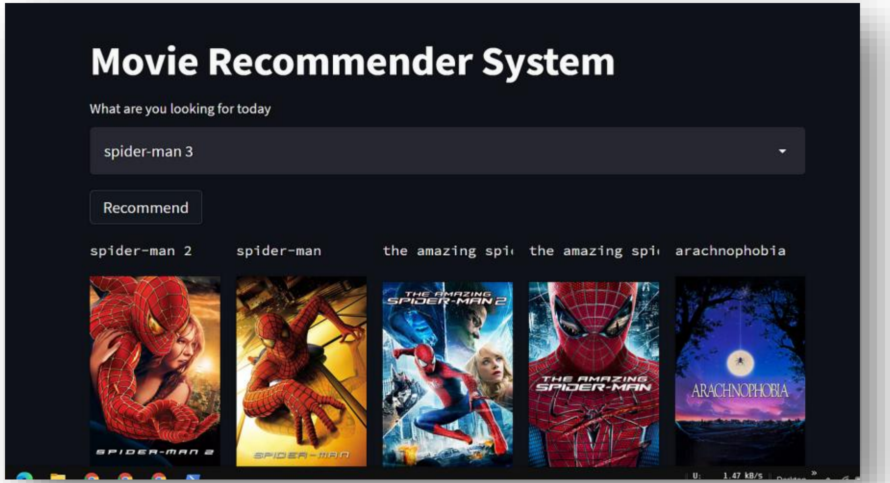
Machine Learning	Deep Learning
• Apply statistical algorithms to learn hidden patterns and relationships in the dataset.	• Uses artificial neural network architecture to learn the hidden patterns and relationships in the dataset.
• Can work on the smaller amount of dataset	• Requires larger volume of dataset
• Better for low-label task.	• Better for complex task like image processing, natural language processing, etc.
• Takes less time to train models.	• Takes more time to train the model.
• A model is created by relevant features which are manually extracted from images to detect an object in the image.	• Relevant features are automatically extracted from images.
• Less complex and easy to interpret the result.	• More complex, it works like a black box and interpretations of the result are not easy.
• It can work on the CPU or requires less computing power as compared to deep learning.	• Requires a high-performance computer.

Deep Learning : Applications

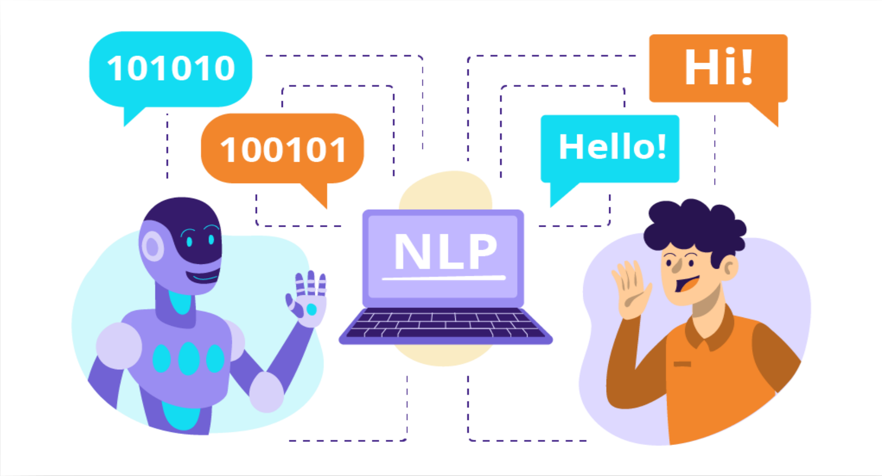
Object Detection



Advanced recommendation systems



Natural Language processing



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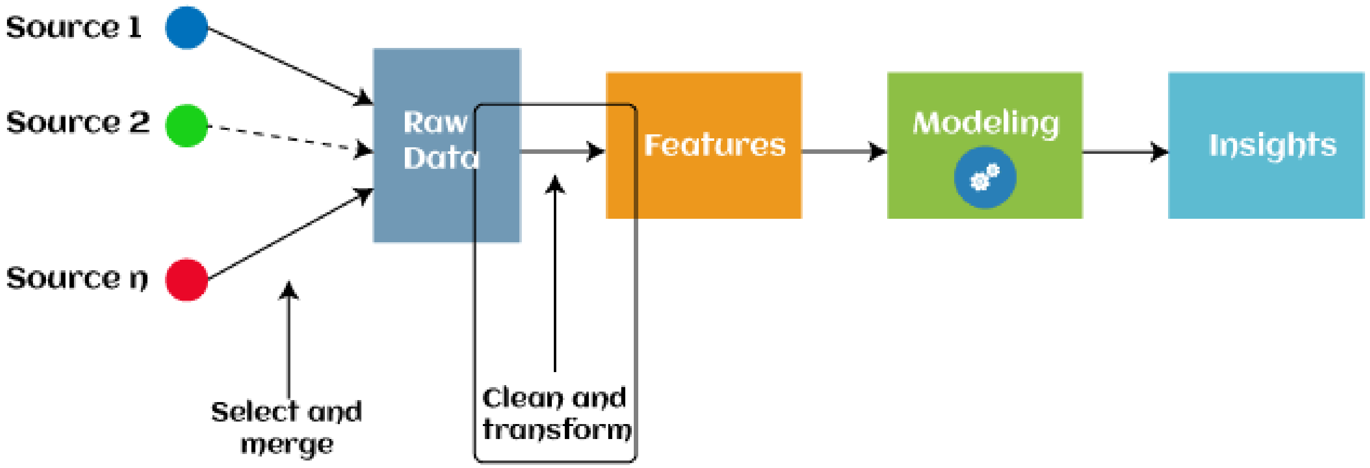
Feature Engineering : Purpose

Why Do Feature Engineering ?

- ✓ Feature engineering involves Cleaning, transforming, selecting features (input variables) and also creating new ones to improve the performance of machine learning models.
- ✓ It's crucial because the quality and relevance of features directly affect the model's ability to learn patterns and make accurate predictions.

Feature Engineering Techniques :

- ✓ Handling Missing Data
- ✓ Outlier Detection
- ✓ Scaling & Normalization
- ✓ Encoding Categorical Data



Feature Engineering : Handling Missing Data

Missing Data : the values or data that is not stored (or not present) for some variables in the given dataset

Missing values

PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	male	22	1	0	A/5 21171	7.25		S
2	1	1	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	female	26	0	0	STON/O2. 3101282	7.925		S
4	1	1	female	35	1	0	113803	53.1	C123	S
5	0	3	male	35	0	0	373450	8.05		S
6	0	3	male		0	0	330877	8.4583		Q

Feature Engineering : Handling Missing Data

Principal methods of handling missing data :

- Handling Missing Data
  - By deleting it
  - By filling it

1) By deleting it

- By deleting it
  - Deleting entire row
  - Deleting entire column

Feature Engineering : Handling missing data

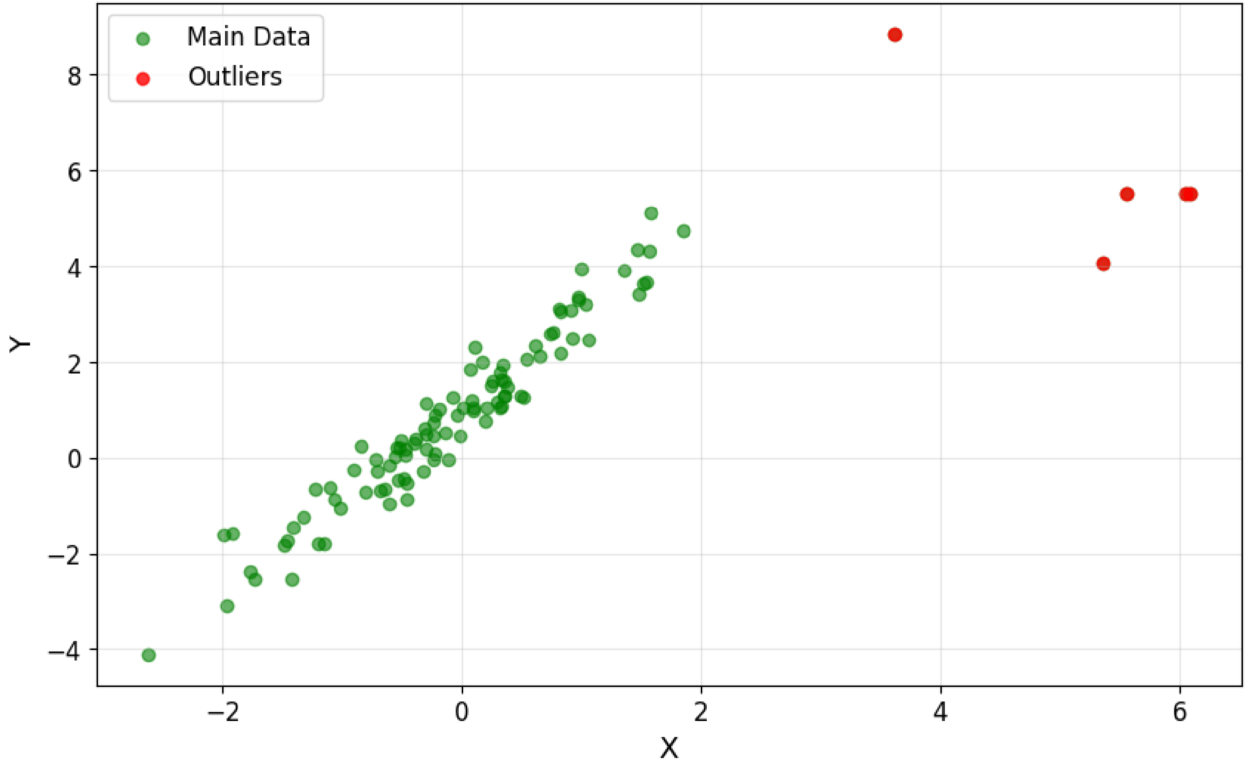
2) By filling it

- By filling it
  - Arbitraty value
  - Mean
  - Mode
  - Median
  - previous value - forward value
  - next value- backward fill
  - Interpolation

Feature Engineering : Outlier Detection

What are Outliers ?

Outliers are the values that look different from the other values in the data.
Below is a plot highlighting the outliers in '**red**' and outliers can be seen in both the extremes of data.



Feature Engineering : Outlier Detection

Reasons for outliers in data

- ➔ Errors during data entry or a faulty measuring device (a faulty sensor may result in extreme readings).
- ➔ Natural occurrence (salaries of junior level employees vs C-level employees)

Problems caused by outliers

- ➔ Outliers in the data may cause problems during model fitting (especially linear models).
- ➔ Outliers may inflate the error metrics which give higher weights to large errors (example: MSE , RMSE).

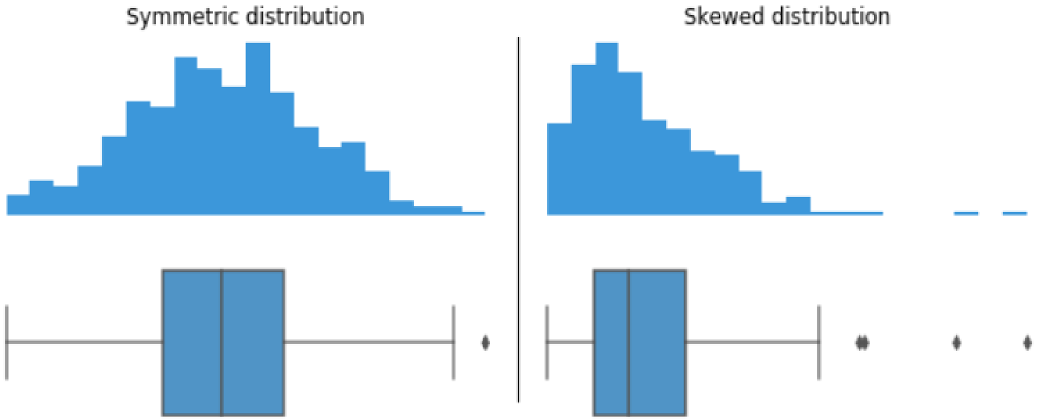
Feature Engineering : Outlier Detection

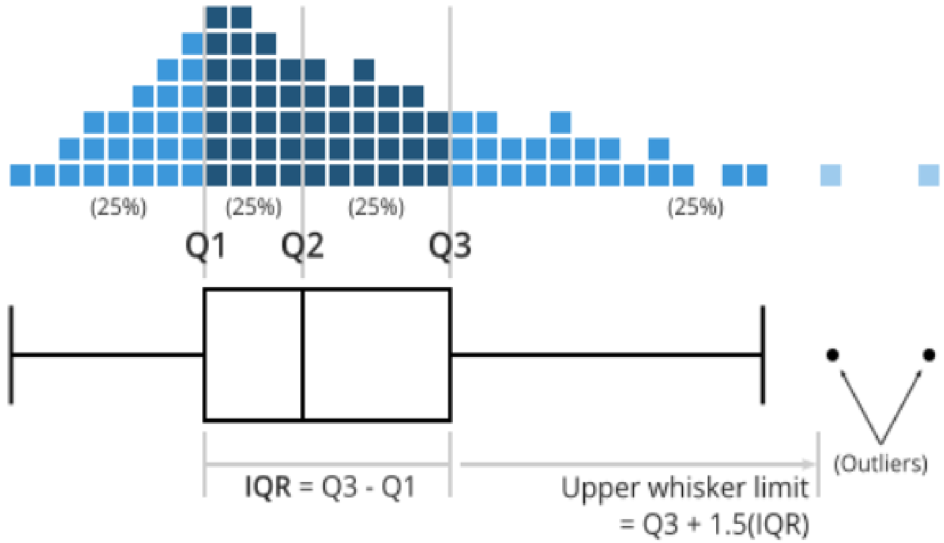
Box plot & IQR method for Outlier Detection

- ➔ A box plot (Box and whisker plot) uses boxes and lines to depict the distributions of one or more groups of numeric data.
- ➔ Box limits indicate the range of the central 50% of the data, with a central line marking the median value.
- ➔ Lines extend from each box to capture the range of the remaining data, with dots placed past the line edges to indicate outliers.

In the presence of outliers :

- ➔ Upper whisker limit = Q3 + 1.5*IQR
- ➔ Lower whisker limit = Q1 - 1.5*IQR





Feature Engineering : Scaling & Normalization

Scaling

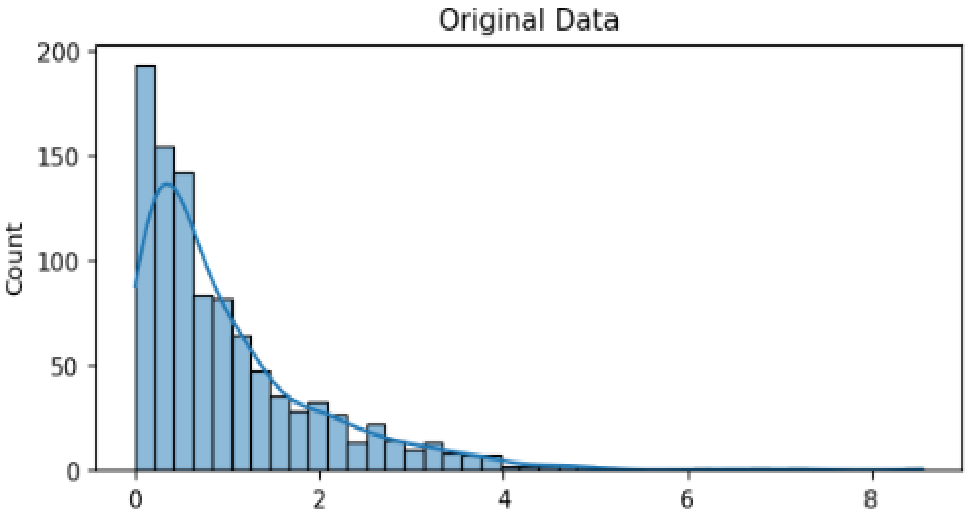
- ✓ This means that you're transforming your data so that it fits within a **specific scale**, like 0-100 or 0-1.
- ✓ You want to scale data when you're using Distance-based algorithms
- ✓ Distance-based algorithms, such as the KNN, calculate the distance between data points to determine their similarity.
- ✓ Features with larger magnitudes can disproportionately influence the distance calculation, leading to biased results.
- ✓ With these algorithms, a change of "1" in any numeric feature is given the same importance.
- ✓ Feature scaling addresses this issue by transforming the features to a comparable range or scale, ensuring that each feature contributes fairly to the final result.

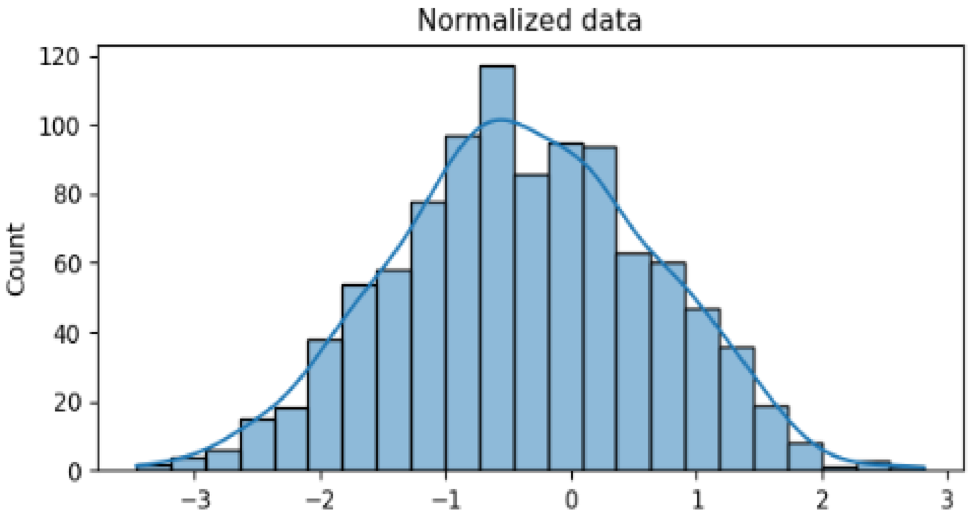
Normalization

- ✓ Scaling just changes the range of your data.
- ✓ **Normalization is a more radical** transformation.
- ✓ The point of normalization is to change your observations so that they can be described as a **normal distribution**.

Normal distribution

- ✓ Normal distribution, also known as the "bell curve",
- ✓ It is a specific statistical distribution where a roughly equal observations fall above and below the mean,
- ✓ The mean and the median are the same,
- ✓ There are more observations closer to the mean.
- ✓ The normal distribution is also known as the Gaussian distribution.





Feature Engineering : Scaling & Normalization - Formulas

Z-score Normalization :

We calculate the mean μ and standard deviation $\boldsymbol{\sigma}$ of the variable. Then for each observation, we subtract the mean and then divide by the standard deviation of that variable

$$z = \frac{x - \mu}{\sigma}$$

Min-Max scaling :

For each observation, we subtract the minimum value and then divide by the difference between the maximum and minimum value

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Result after normalization

ID	Age	Income(rupees)
1	-1.192	-0.260
2	-0.447	0.608
3	1.043	0.173
4	-0.447	-1.563
5	1.043	1.042

Feature Engineering : Scaling & Normalization – Example

Example

- In the table, we observe that the Age of the person ranges from 25 to 40 whereas the income variable ranges from 50,000 to 110,000.
- when we calculate the euclidean distance between point 1 and 2, the result is 20000.000625.
- The high magnitude of income affected the distance between the two points. T
- To overcome this problem, we can bring down all the variables to the same scale using z-score normalization and Min-Max scaling.

ID	Age	Income(rupees)
1	25	80,000
2	30	100,000
3	40	90,000
4	30	50,000
5	40	110,000

Feature Engineering : Scaling & Normalization

Example

1) z-score Normalization

To apply normalization, we need to calculate mean and deviation.

Mean $$\mu = \frac{\sum_{i=1}^{n} x_i}{n}$$

Deviation $$\sigma = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \mu)^2}{n}}$$

In python:

```
# Preprocess numerical features (scaling)
numeric_transformer = Pipeline(steps=[('scaler', StandardScaler())])
```

ID	Age	weight (kg)	salary ($)
1	30	70	50000
2	40	80	60000
3	50	90	70000

Feature Engineering : Scaling & Normalization

Calculate Mean:

- For Age: $\mu_{\text{Age}} = \frac{30+40+50}{3} = \frac{120}{3} = 40$
- For Weight: $\mu_{\text{Weight}} = \frac{70+80+90}{3} = \frac{240}{3} = 80$
- For Salary: $\mu_{\text{Salary}} = \frac{50000+60000+70000}{3} = \frac{180000}{3} = 60000$

Calculate Standard Deviation:

- For Age:

$$\sigma_{\text{Age}} = \sqrt{\frac{(30-40)^2+(40-40)^2+(50-40)^2}{3}} = \sqrt{\frac{100+0+100}{3}} = \sqrt{\frac{200}{3}} \approx 8.16$$

- For Weight:

$$\sigma_{\text{Weight}} = \sqrt{\frac{(70-80)^2+(80-80)^2+(90-80)^2}{3}} = \sqrt{\frac{100+0+100}{3}} = \sqrt{\frac{200}{3}} \approx 8.16$$

- For Salary:

$$\sigma_{\text{Salary}} = \sqrt{\frac{(50000-60000)^2+(60000-60000)^2+(70000-60000)^2}{3}} = \sqrt{\frac{1000000+0+1000000}{3}} = \sqrt{\frac{2000000}{3}} \approx 816.50$$

Result after normalization

ID	Normalized Âge	Normalized weight (kg)	Normalized salary ($)
1	-1,22	-1,22	-1,22
2	0	0	0
3	1,22	1,22	1,22

✓ For each value of the data frame, we subtract the mean of its corresponding feature and divide by the deviation

Feature Engineering : Scaling & Normalization

2) Min – Max Scaling $$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Find Min and Max Values:

- For Age: $\mathrm{Min}_{\mathrm{Age}} = 30, \mathrm{Max}_{\mathrm{Age}} = 50$
- For Weight: $\mathrm{Min}_{\mathrm{Weight}} = 70, \mathrm{Max}_{\mathrm{Weight}} = 90$
- For Salary: $\mathrm{Min}_{\mathrm{Salary}} = 50000, \mathrm{Max}_{\mathrm{Salary}} = 70000$

The application of the first point

- For Age: $\frac{30-30}{50-30} = \frac{0}{20} = 0$
- For Weight: $\frac{70-70}{90-70} = \frac{0}{20} = 0$
- For Salary: $\frac{50000-50000}{70000-50000} = \frac{0}{20000} = 0$

Result after Min-Max scaling

IDc	Normalized Âge	Normalized weight (kg)	Normalized salary ($)
1	0	0	0
2	0,5	0,5	0,5
3	1	1	1

In python:

```
from sklearn.preprocessing import MinMaxScaler

numeric_transformer = Pipeline(steps=[('scaler', MinMaxScaler())])
```

Feature Engineering : Encoding Categorical Data

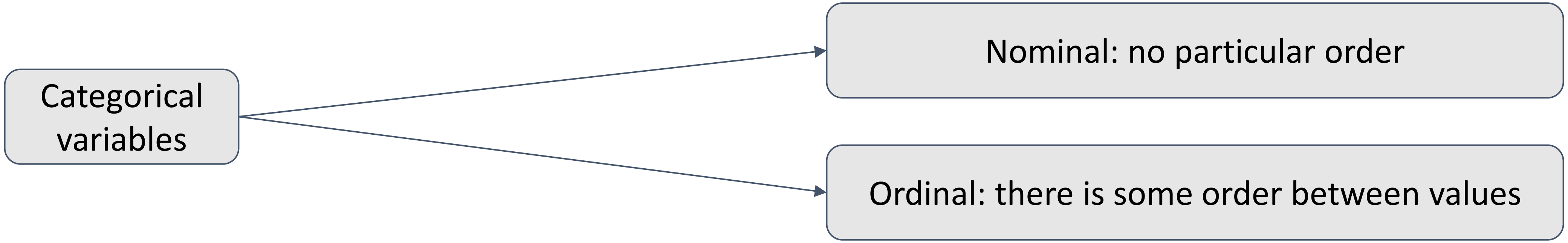
Categorical Vs Numerical Data :

- ✓ Categorical data is a type of data that is used to group information with similar characteristics, while numerical data is a type of data that expresses information in the form of numbers.
- ✓ Example of categorical data: **Gender**

Why do we need encoding Categorical Data?

- ✓ Most machine learning algorithms cannot handle categorical variables unless we convert them to **numerical values**

Categorical variables can be divided into two categories :



Method 1: Using Python’s Category Encoders Library

category_encoders is an amazing Python library that provides 15 different encoding schemes. it supports 15 types of encoding such as : One-hot encoding, Label encoding, Ordinal Encoding, Binary encoding …etc.

Python application

Importing libraries:

```
import pandas as pd
import sklearn

pip install category_encoders

import category_encoders as ce
```

Feature Engineering : Encoding Categorical data

Creating a dataframe:

```
data = pd.DataFrame({ 'gender' : ['Male', 'Female', 'Male', 'Female', 'Female'],
                      'class' : ['A','B','C','D','A'],
                       'city' : ['Delhi','Gurugram','Delhi','Delhi','Gurugram'] })
data.head()
```

	gender	class	city
0	Male	A	Delhi
1	Female	B	Gurugram
2	Male	C	Delhi
3	Female	D	Delhi
4	Female	A	Gurugram

Feature Engineering : Encoding Categorical data

Implementing **One-Hot Encoding** through category_encoder

```
ce_OHE = ce.OneHotEncoder(cols=['gender','city'])

data1 = ce_OHE.fit_transform(data)
data1.head()
```

	gender_1	gender_2	class	city_1	city_2
0	1	0	A	1	0
1	0	1	B	0	1
2	1	0	C	1	0
3	0	1	D	1	0
4	0	1	A	0	1

	gender	class	city
0	Male	A	Delhi
1	Female	B	Gurugram
2	Male	C	Delhi
3	Female	D	Delhi
4	Female	A	Gurugram

✓ Similarly, there are another 14 types of encoding provided by this library

Feature Engineering : Encoding Categorical data

Method 2: Using Pandas' Get Dummies

```
pd.get_dummies(data,prefix=["gen","city"],columns=["gender","city"])
```

	class	gen_Female	gen_Male	city_Delhi	city_Gurugram
0	A	0	1	1	0
1	B	1	0	0	1
2	C	0	1	1	0
3	D	1	0	1	0
4	A	1	0	0	1

Feature Engineering : Encoding Categorical data

Method 3: Using Scikit-learn Ordinal Encoding

First, we need to assign the original order of the variable through a dictionary.

```
temp = {'temperature' :['very cold', 'cold', 'warm', 'hot', 'very hot']}
df=pd.DataFrame(temp,columns=["temperature"])
temp_dict = {'very cold': 1,'cold': 2,'warm': 3,'hot': 4,"very hot":5}
df
```

	temperature
0	very cold
1	cold
2	warm
3	hot
4	very hot

Then we can map each row for the variable as per the dictionary.

```
df["temp_ordinal"] = df.temperature.map(temp_dict)
df
```

Final Result :

	temperature	temp_ordinal
0	very cold	1
1	cold	2
2	warm	3
3	hot	4
4	very hot	5

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Lab 2: Improve your model with Feature Engineering

Part 1 : Models comparison

- ➔ Go back to lab 1 and apply the metrics we have seen.
- ➔ Based on the metrics, select the best model.

Part2 : Select one of the models, ideally Random Forest, to improve its performance.

- ➔ Load Spotify dataset SpotifyDataset_WithNA.csv
- ➔ Define the feature and the target
- ➔ Calculate the number of missing values and drop the rows containing missing values
- ➔ Split the dataset into training and test set
- ➔ Preprocess numerical and categorical features.
- ➔ Train & Fit the selected model
- ➔ Predict on Test sets
- ➔ Calculate metrics

Thank you!

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